DATA MINING AND MACHINE LEARNING

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS16	PCC	3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Learn Data Mining concepts, preprocessing techniques
2. Understand associate rules for frequent pattern mining and learn concept of supervised and un supervised learning techniques
3. Discuss various classification algorithms.
4. Befamiliarwithbasicmachinelearningalgorithmswithclassificationandclustering.
5. Gain experience of doing in dependent study and research.

COURSEOUTCOMES

Upon successful completion of this course, student will be able to:

1. Perform the preprocessing of data and apply mining techniques on it.
2. Recognize the characteristics of machine learning that makes it useful to solve real-world problems.
3. Design and implement machine learning solutions of classification, regression problems.
4. Choose an appropriate clustering technique to solve real world problems.
5. Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real world problems.

UNIT- I	INTRODUCTION	CLASSES: 12

INTRODUCTION TO DATA MINING: Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values.
PREPROCESSING: Data Quality, Major Tasks in Data Pre-processing, Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization, Measures of Similarity and Dissimilarity-Basics

UNIT - II	ASSOCIATION RULES	CLASSES: 12

INTRODUCTION TO MACHINE LEARNING: Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks.
ASSOCIATION RULES: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIOIRI Algorithm, Compact Representation of Frequent Item Set- Maximal Frequent Item Set

UNIT - III	SUPERVISED LEARNING	CLASSES: 14

Learning a Class from Examples, Linear, Non-linear, Multi-class and Multi-label classification, Perceptron: – multilayer neural networks – back propagation - learning neural networks structures –support vector machines:–soft margin SVM–going beyond linearity generalization and over fitting– regularization— validation.
Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.

UNIT - IV	UNSUPERVISED LEARNING	CLASSES: 14

Introduction to clustering, Hierarchical: AGNES,DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis,
Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and Non Linear, Kernel Functions, K-Nearest Neighbors.

UNIT - V	ENSEMBLE AND PROBABILISTIC LEARNING	CLASSES: 12

Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking, Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesi an Belief Networks, Mining Frequent Patterns

TEXT BOOKS

1. Data Mining: Concepts and Techniques – Jiawei Han, Micheline Kamber, Jian Pei (2012), 3^{rd} edition, Elsevier, United States of America.
2. Giuseppe Bonaccorso — Machine Learning Algorithms ‖, 2^{nd} Edition, Packt, 2018.

REFERENCE BOOKS

1. Data Mining Techniques, Arun K Pujari, 3rd Edition, Universities Press.
2. Data Mining Principles & Applications–T.V S veresh Kumar, B. Esware Reddy, Jagadish S Kalimani, Elsevier.
3. Ethem Al paydin,—Introduction to Machine Learning ‖,PHI, 2004
4. Stephen Marshland, —Machine Learning: An Algorithmic Perspective‖, CRC Press Taylor & Francis, 2nd Edition, 2015